Assessing Public Transit Accessibility in Philadelphia and Predicting Underserved Neighborhoods Using PySpark

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Abstract

Public transportation accessibility plays a crucial role in urban mobility, economic opportunity, and social equity. Cities with uneven transit distribution often experience "transit deserts," areas where residents face limited access to reliable transportation. This study evaluates transit accessibility across 159 neighborhoods in Philadelphia using SEPTA General Transit Feed Specification (GTFS) data combined with neighborhood boundary datasets. The project processes more than 2.7 million transit schedule records using PySpark to enable scalable distributed computation. Ten spatial and service-based features are engineered for each neighborhood, including stop density, service frequency, route diversity, and proximity to transit infrastructure. A composite accessibility score is computed to quantify neighborhood-level transit access. Machine learning models, including Random Forest and Gradient Boosted Trees implemented in PySpark MLlib, are used to predict accessibility scores and identify underserved areas. The Random Forest model achieved an R^2 value of 0.971, demonstrating strong predictive performance. Results reveal substantial disparities in transit access across Philadelphia, with central neighborhoods having significantly higher accessibility scores than peripheral regions. The findings demonstrate how large-scale transit data and distributed computing techniques can support data-driven urban planning decisions.

I. Introduction

A. Motivation

Public transportation is a critical component of urban infrastructure. It provides access to employment opportunities, education, healthcare, and other essential services. Cities with equitable transit systems enable greater economic mobility and improved quality of life for residents.
However, transit networks are often unevenly distributed. In many cities, including Philadelphia, certain neighborhoods benefit from dense transit networks while others have limited connectivity. These underserved regions are commonly referred to as transit deserts, where residents face difficulty accessing reliable transportation options.
Philadelphia's transit system is operated by the Southeastern Pennsylvania Transportation Authority (SEPTA), which manages buses, subways, trolleys, and regional rail services. Although SEPTA operates one of the largest transit networks in the United States, the accessibility of this system varies significantly across neighborhoods.
Understanding these disparities is essential for policymakers and urban planners who seek to improve transportation equity. Identifying areas with limited transit access allows agencies to prioritize infrastructure investments, adjust service frequency, and improve connectivity for underserved communities.

B. Research Questions

This project seeks to answer the following research questions:

1. Which Philadelphia neighborhoods experience the lowest levels of transit accessibility?
2. Which transit-related factors most strongly influence accessibility?
3. Can machine learning models accurately predict accessibility scores based on spatial and service features?

C. Contributions

The primary contributions of this work include:

- A scalable PySpark pipeline for processing large transit datasets.

- A feature engineering framework for measuring neighborhood-level transit accessibility.
- A composite accessibility scoring model combining spatial and service indicators.
- Machine learning models capable of predicting underserved transit areas.

II. Infrastructure and Tools

The analysis pipeline was implemented using distributed data processing tools to handle the large volume of transit schedule data.

A. Technology Stack

This project utilized many libraries and technologies to produce the best result.

Category	Tools
Programming	Python 3.x, PySpark 3.5.0
Cloud Platform	AWS EMR, AWS S3
Development	Google Colab
Machine Learning	PySpark MLlib
Visualization	Folium, Matplotlib, Seaborn
Geospatial	GeoPandas, Shapely

B. Distributed Computing Architecture

The project leverages Apache Spark's distributed computing model to process millions of transit records efficiently. The architecture includes:

- **Driver Program:** Coordinates the Spark application and manages task scheduling.
- **Worker Nodes:** Execute distributed tasks across partitions of the dataset.
- **Executors:** Perform computations on each partition of data.

This architecture enables large datasets to be processed in parallel, significantly reducing computation time.

III. Dataset Description

A. Data Sources

Two primary datasets were used in this study.

1. SEPTA GTFS Transit Data

The General Transit Feed Specification (GTFS) dataset provides detailed information about transit stops, routes, schedules, and trip times.

Key files include:

Table. I. Number of rows and description in each dataset

File	Rows	Description
stop_times.txt	2,733,692	Transit arrival and departure times
trips.txt	46,202	Trip definitions linking routes to schedules
routes.txt	182	Transit route definitions
stops.txt	13,855	Geographic coordinates of transit stops

2. Philadelphia Neighborhood Boundaries

Neighborhood boundary data was obtained from OpenDataPhilly in GeoJSON format. The dataset contains 159 polygon boundaries representing neighborhoods across the city.These polygons were used to assign spatial context to transit accessibility measurements.

IV. Data Exploration and Preparation

A. Data Processing Pipeline

The data processing workflow consisted of several steps:

1. Download SEPTA GTFS data.
2. Extract and load CSV files into PySpark DataFrames.
3. Merge bus and rail stop datasets.
4. Load neighborhood boundary polygons using GeoPandas.
5. Compute geographic centroids for each neighborhood.
6. Calculate spatial distances between neighborhoods and transit stops.

```
=== Bus Stops Sample ===
+-------+--------------------------------+---------+----------+
|stop_id|stop_name                       |stop_lat |stop_lon  |
+-------+--------------------------------+---------+----------+
|4      |Roosevelt Blvd & Broad St - FS  |40.018128|-75.148887|
|5      |Roosevelt Blvd & 5th St         |40.023465|-75.133487|
|7      |Roosevelt Blvd & Pratt St       |40.028754|-75.089524|
|8      |Roosevelt Blvd & Cottman Av - FS|40.04461 |-75.053657|
|9      |Roosevelt Blvd & Holme Av       |40.058496|-75.044312|
+-------+--------------------------------+---------+----------+
```

Fig. 1. Sample records from the SEPTA GTFS datasets showing stops, stop times, and route structures.

B. Exploratory Data Analysis

Initial exploratory analysis revealed the following statistics:

Table. II. Distribution of transit stops in the dataset.

Metric	Value
Total Transit Stops	13,855
Bus Stops	13,699
Rail Stations	156
Bus Routes	169
Rail Lines	13

Metric	Value
Neighborhoods Analyzed	159

Bus stops account for approximately 98.9% of all transit stops, indicating that bus service forms the backbone of Philadelphia's transit network.
To understand the spatial distribution of transit infrastructure, a heatmap of SEPTA stop locations was generated across the Philadelphia region.

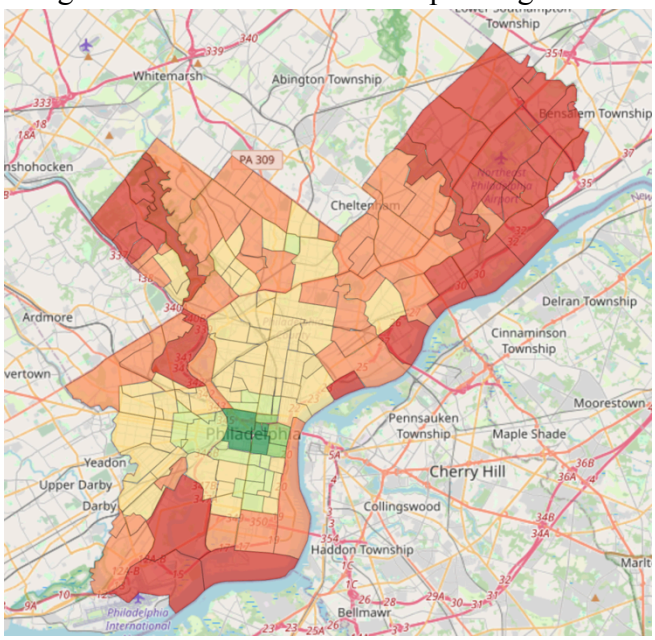

Fig. 2. Heatmap visualization of SEPTA transit stop density across Philadelphia.

V. Feature Engineering

A. Spatial Feature Computation

Feature engineering focused on measuring accessibility using spatial relationships between neighborhoods and transit stops.
The workflow included:

1. Calculating neighborhood centroids.
2. Performing a cross join between neighborhoods and stops.
3. Computing Haversine distances between centroids and stops.
4. Aggregating transit features within distance thresholds.

The cross join produced approximately 2.2 million distance calculations.

B. Engineered Features

Ten features were generated for each neighborhood.

Table. III. LIST OF GENERATED FEATURES

Feature	Description
stops_within_500m	Transit stops within 500 meters
stops_within_1km	Transit stops within 1 kilometer
stops_within_1500m	Transit stops within 1.5 kilometers
min_distance_to_stop_km	Distance to the nearest stop
bus_stops_within_1km	Bus stops within 1 km
rail_stops_within_1km	Rail stations within 1 km
unique_routes_within_1km	Distinct transit routes
total_trips_within_1km	Total daily trips
avg_trips_per_stop	Average service frequency
distance_to_center_city_km	Distance to Philadelphia City Hall

C. Accessibility Score Calculation

A composite accessibility score was computed using a weighted combination of normalized features.

$$Accessibility = 0.25S + 0.25R + 0.25T + 0.15($$

Where:

- **S** = Stop density
- **R** = Route diversity
- **T** = Trip frequency
- **C** = Proximity to Center City
- **P** = Distance to nearest stop

The score ranges from 0 to 1, where higher values indicate better transit accessibility.

VI. Machine Learning Approach

A. Modeling Strategy

Two supervised regression models were implemented:

1. Random Forest Regressor
2. Gradient Boosted Trees Regressor

These models were selected due to their ability to capture nonlinear relationships between features.

B. Preprocessing Pipeline

The machine learning pipeline included:

1. **VectorAssembler** – Combine features into a vector.

2. **StandardScaler** – Normalize feature values.
3. **Train-Test Split** – 80% training, 20% testing.

VII. Model Evaluation and Results

A. Random Forest Performance

Metric	Value
RMSE	0.0263
MAE	0.0202
R^2	0.9711

The Random Forest model explains 97.1% of the variance in accessibility scores.

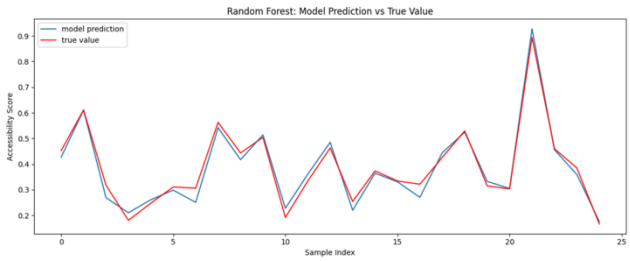

Fig. 3. Comparison of predicted and true accessibility scores across samples.

B. Gradient Boosted Trees Performance

Metric	Value
RMSE	0.0376
MAE	0.0294
R^2	0.9410

Although GBT performed well, Random Forest achieved better predictive accuracy.

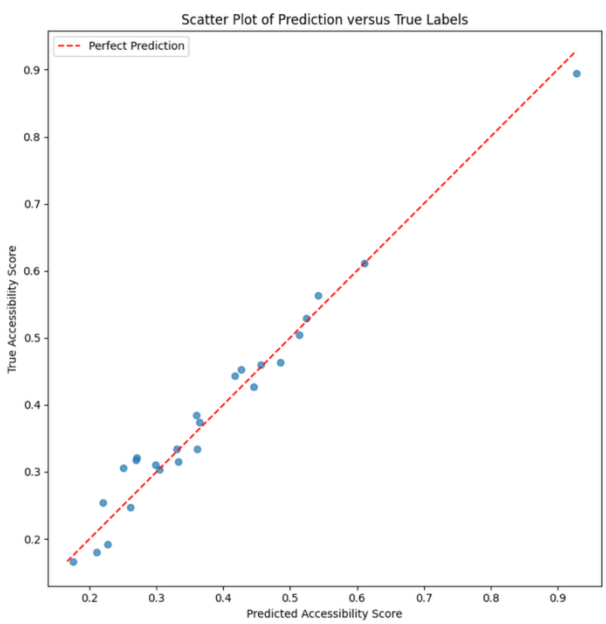

Fig. 4. Scatter plot comparing predicted and actual accessibility scores.

C. Feature Importance

Feature importance analysis revealed that:

- Stops within 1 km – **23.3%**
- Bus stops within 1 km – **19.5%**
- Total trips within 1 km – **19.5%**
- Stops within 1.5 km – **14.6%**

These findings indicate that stop density and service frequency are the strongest drivers of accessibility. Accessibility scores were visualized geographically using a choropleth map to highlight spatial disparities in transit coverage.

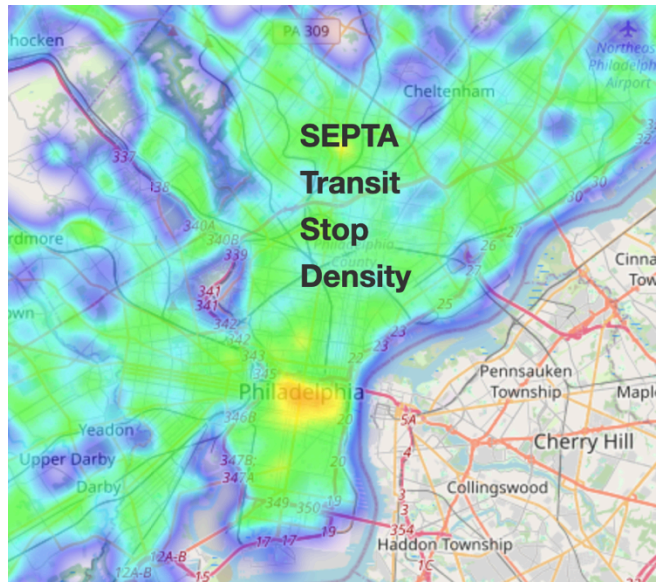

Fig. 5. Geographic distribution of transit accessibility scores across Philadelphia neighborhoods

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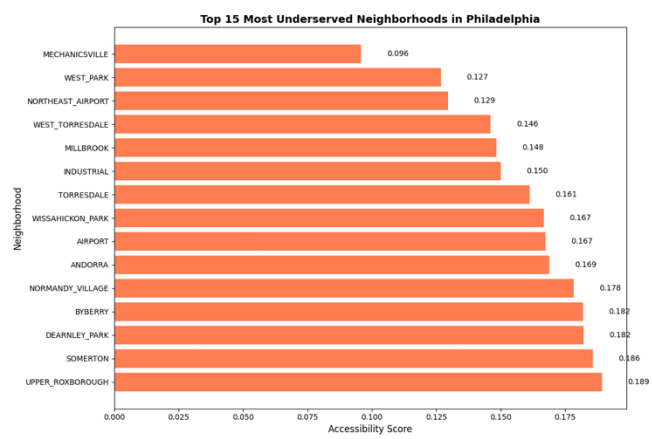

Fig. 6. Top 15 most underserved neighborhoods ranked by accessibility score.

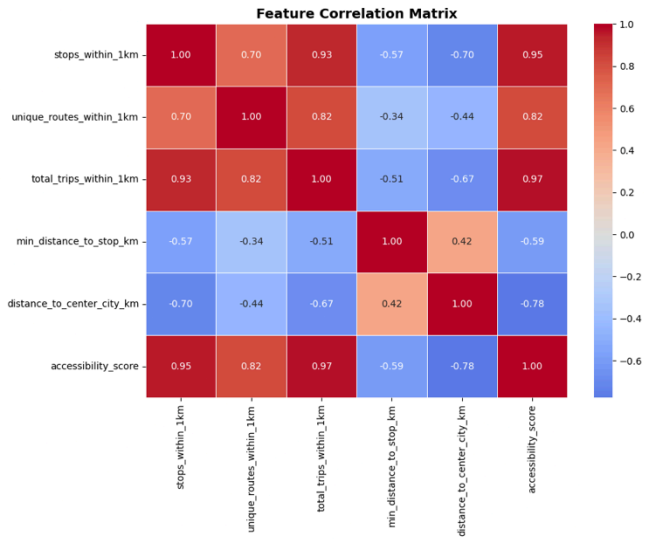

Fig. 7. Feature correlation matrix heatmap.

VIII. Visualization

Several visualizations were generated to communicate findings:

1. Choropleth map of accessibility scores
2. Transit stop density heatmap
3. Underserved neighborhood ranking
4. Feature correlation matrix
5. Model prediction comparison plots

These visualizations highlight spatial disparities in transit access.

IX. AWS EMR Deployment

To demonstrate real-world cloud scalability, the pipeline was deployed on Amazon EMR.

Cluster Configuration

Component	Configuration
EMR Version	emr-7.0.0
Master Node	m5.xlarge
Core Nodes	2 × m5.large
Applications	Spark, Hadoop
Storage	Amazon S3

The architecture allows the analysis pipeline to scale for larger transit datasets or multi-city comparisons.

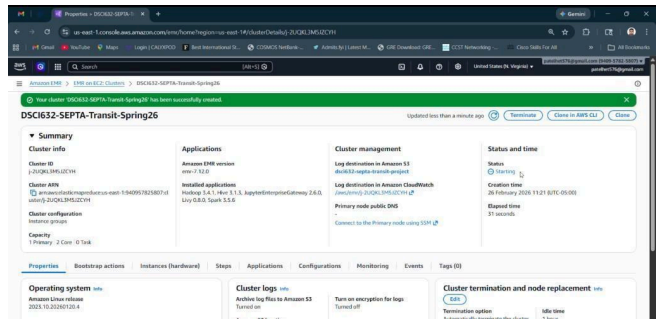

Fig. 8. AWS EMR cluster successfully created with Spark 3.5.6, Hadoop 3.4.1, and JupyterEnterpriseGateway.

X. Limitations

Several limitations should be considered:

Centroid Approximation

Distances were calculated from neighborhood centroids rather than full polygon boundaries, which may introduce spatial approximation errors.

Straight-Line Distance

Haversine distance measures straight-line travel rather than actual walking routes along street networks.

Temporal Aggregation

Transit schedules were aggregated rather than analyzed by time of day, meaning peak and off-peak accessibility differences were not captured.

Aws Development

While the EMR cluster and S3 integration were successfully configured, the cluster steps failed to initialize due to dependencies of libraries (e.g., folium) that were available in the local Jupyter environment but properly captured in the EMR runtime environment.

XI. Conclusion and Future Work

A. Conclusion

This study demonstrates how distributed data processing and machine learning can be applied to evaluate transit accessibility in urban environments.

Key findings include:

- Large disparities exist between central and peripheral neighborhoods.
- Stop density and service frequency are the strongest predictors of transit accessibility.

- Random Forest models provide highly accurate predictions of accessibility scores.

These results provide actionable insights for improving transit equity across Philadelphia.

B. Future Work

Several extensions could further improve this analysis.

Time-Based Routing

Future work could incorporate time-dependent routing analysis using detailed transit schedules from the GTFS dataset. Rather than evaluating accessibility at the neighborhood level alone, the system could allow users to select an origin and destination and compute the optimal transit path between them.

By using arrival and departure information from the stop_times.txt dataset, it would be possible to estimate travel times between locations at different times of day. For example, the system could calculate the fastest route between two neighborhoods during morning rush hour compared to late evening service hours.

Origin–Destination Accessibility Modeling

Another extension involves computing relative accessibility scores between neighborhoods. This would measure how easily residents from one neighborhood can reach another neighborhood or key destinations such as employment centers, hospitals, or universities.

Such models could incorporate graph-based shortest path algorithms, such as Dijkstra's algorithm, applied to the transit network.

Additional Data Integration

Further improvements may include:

- Integration of census demographic data to analyze transit equity
- Incorporation of real-time service delays
- Network-based walking distance calculations using road network graphs
- Expansion of the framework to multiple cities

These improvements would transform the current system into a dynamic transit planning and decision-support tool capable of evaluating real-world travel scenarios.

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